Numerical Solution of the Time-Dependent Schrödinger Equation via the Split-Step Fourier Method

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Abstract

We implement and validate a split-step Fourier method (SSFM) solver for the time-dependent Schrödinger equation (TDSE) in one and two dimensions. The method uses the fact that the kinetic-energy operator $\hat{T}$ is diagonal in momentum space and the potential operator $\hat{V}$ is diagonal in position space, and moves between the two with the fast Fourier transform (FFT). Each step is a product of unitary operations, so the norm of the wavefunction is conserved to machine precision; with symmetric (Strang) splitting the scheme is second-order accurate in the time step Δt. We test the solver against five problems with known answers: (1) free Gaussian dispersion, where the width matches the exact result to 10^{-14} once the box is large enough; (2) rectangular-barrier tunnelling, where transmission agrees with the momentum-averaged exact result to within 5×10^{-4} over 15 energies; (3) a harmonic-oscillator coherent state, whose centre follows the classical orbit to $1.4 \times 10^{-4}\, a_0$ and whose error falls as $\Delta t^{2.000}$; (4) two-dimensional double-slit interference, where the fringes sit at $k_y = 2\pi m/d$ to within one momentum-grid bin; and (5) partial reflection from a potential step, where R agrees with theory to within 8×10^{-5}. The solver is short, fast and accurate, and provides a base for future work on laser–matter interaction and open quantum systems.

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1 Motivation and physical background

1.1 Why solve the TDSE numerically?

The time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = \hat{H}\psi(x,t) = \big[\hat{T} + \hat{V}(x)\big]\psi(x,t), \tag{1}$$

governs the non-relativistic dynamics of a quantum particle. Here $\hat{H}$ is the Hamiltonian, $\hat{T} = -(\hbar^2/2m)\,\partial^2/\partial x^2$ is the kinetic-energy operator, and $\hat{V}(x)$ is the potential, which acts by multiplication in position space.

Closed-form solutions of (1) exist only for a handful of potentials, such as the free particle, the harmonic oscillator and the step and square barrier. For realistic problems (an atom in a laser field, tunnelling through a shaped barrier, a wavepacket scattering off a defect) numerical integration is the only option.

The split-step Fourier method is a standard tool for this in quantum optics and atomic, molecular and optical physics [3, 6]. Each step costs $O(N \log N)$ operations, it conserves the norm exactly, and it takes a few lines of code.

1.2 What is the wavefunction?

The wavefunction $\psi(x,t)$ is complex. Its physical content is in $|\psi(x,t)|^2$, the probability density of finding the particle at x at time t. Normalisation requires

$$\int_{-\infty}^{\infty} |\psi(x,t)|^2\, dx = 1 \quad \text{for all } t. \tag{2}$$

The TDSE preserves this: if $\psi(x,0)$ is normalised, so is $\psi(x,t)$ at every later time. A numerical method that violates this is wrong.

1.3 Atomic units

We use atomic units,

$$\hbar = m_e = e = a_0 = 1, \tag{3}$$

where $a_0 \approx 0.529\,\text{Å}$ is the Bohr radius and the unit of energy is the Hartree, $E_h \approx 27.2\,\text{eV}$. The TDSE becomes

$$i\,\frac{\partial \psi(x,t)}{\partial t} = \left[-\frac{1}{2}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x,t). \tag{4}$$

All results below are in atomic units.

2 The split-step Fourier method

2.1 The key difficulty: $\hat{T}$ and $\hat{V}$ do not commute

The formal solution over one time step is

$$\psi(x, t+\Delta t) = e^{-i\hat{H}\Delta t/\hbar}\psi(x,t) = e^{-i(\hat{T}+\hat{V})\Delta t/\hbar}\psi(x,t). \tag{5}$$

If $\hat{T}$ and $\hat{V}$ commuted we could write this as $e^{-i\hat{T}\Delta t/\hbar}e^{-i\hat{V}\Delta t/\hbar}$. In general $[\hat{T},\hat{V}] \neq 0$, and the Baker–Campbell–Hausdorff (BCH) formula gives the cost:

$$e^{A+B} = e^{A}e^{B}e^{-\frac{1}{2}[A,B]+\cdots}. \tag{6}$$

With $A = -i\hat{T}\Delta t/\hbar$ and $B = -i\hat{V}\Delta t/\hbar$ the leading correction is $O(\Delta t^2)$ per step, so this simple splitting is first-order accurate overall. The symmetric (Strang) splitting [4] cancels the second-order term:

$$e^{-i(\hat{T}+\hat{V})\Delta t/\hbar} = e^{-i\hat{V}\Delta t/2\hbar}\, e^{-i\hat{T}\Delta t/\hbar}\, e^{-i\hat{V}\Delta t/2\hbar} + O(\Delta t^3). \tag{7}$$

The error per step is $O(\Delta t^3)$. Over the $\sim T/\Delta t$ steps of a run it accumulates to $O(\Delta t^2)$, so the scheme is second-order. Section 3.3 confirms this order numerically.

2.2 Why each operator is easy to apply on its own

Potential step. In position space $\hat{V}$ multiplies: $\hat{V}\psi(x) = V(x)\psi(x)$. So

$$e^{-i\hat{V}\Delta t/2\hbar}\psi(x) = e^{-iV(x)\Delta t/2\hbar}\,\psi(x), \tag{8}$$

a pointwise multiplication of two arrays.

Kinetic step. In momentum space $\hat{T}$ also multiplies. The Fourier transform of $\partial^2\psi/\partial x^2$ is $-k^2\tilde{\psi}(k)$, so

$$\hat{T}\tilde{\psi}(k) = \frac{\hbar^2 k^2}{2m}\tilde{\psi}(k), \qquad e^{-i\hat{T}\Delta t/\hbar}\tilde{\psi}(k) = e^{-i\hbar k^2\Delta t/2m}\,\tilde{\psi}(k). \tag{9}$$

2.3 The complete algorithm

One SSFM step is:

1. Half kick in x-space: $\psi \leftarrow \psi \cdot \exp\big(-iV\Delta t/2\hbar\big)$.
2. FFT to k-space: $\tilde{\psi} \leftarrow \mathcal{F}[\psi]$.
3. Full kick in k-space: $\tilde{\psi} \leftarrow \tilde{\psi} \cdot \exp\big(-i\hbar k^2\Delta t/2m\big)$.
4. Inverse FFT to x-space: $\psi \leftarrow \mathcal{F}^{-1}[\tilde{\psi}]$.
5. Half kick in x-space: $\psi \leftarrow \psi \cdot \exp\big(-iV\Delta t/2\hbar\big)$.

For real V every factor has unit modulus, and the discrete Fourier transform is unitary up to a constant. Each step is therefore unitary, and $\int |\psi|^2 dx$ is conserved to round-off. In two dimensions the same steps apply, with $k^2 \to k_x^2 + k_y^2$ and a 2D FFT.

A useful special case: for $V = 0$ the kinetic propagator is exact, so the free-particle runs in Section 3.1 carry no time-stepping error at all. They test the spatial grid and the diagnostics. The time-step error is measured in Section 3.3, where $V \neq 0$.

2.4 Grid, diagnostics and boundaries

The interval $[-L, L)$ is divided into N points with spacing $\Delta x = 2L/N$. The largest resolvable wavenumber is $k_{\max} = \pi/\Delta x$, which must exceed every wavenumber the wavepacket populates. Norms and moments are computed with the rectangle rule, $\int f\,dx \approx \Delta x \sum_j f_j$, which is spectrally accurate for smooth functions on a periodic grid.

The FFT makes the domain periodic, so anything leaving one edge re-enters at the other. We either make the box large enough that this never matters, or add an absorbing layer: a negative imaginary potential $-iW(x)$ near the edges, which damps outgoing waves instead of reflecting them [5]. Table 1 lists the parameters of every run.

Simulation	N	box	Δx	Δt	packet (x_0, σ_0, k_0)	boundary
1 Free particle	1024 / 2048	$[-20, 20)$ / $[-40, 40)$	0.039	0.01	$(-8,\ 1,\ 5)$	periodic
2 Barrier	16384	$[-256, 256)$	1/32	0.02	$(-100,\ 10,\ 0.6\text{–}2.0)$	periodic
3 Oscillator	1024	$[-20, 20)$	0.039	0.01	$(3,\ 1/\sqrt{2},\ 0)$	periodic
4 Double slit	1024×512	$[-20, 60) \times [-30, 30)$	0.078, 0.117	0.004	see §3.4	absorbing
5 Step	16384	$[-300, 300)$	0.037	0.01	$(-100,\ 10,\ 1.6\text{–}3.0)$	periodic

Table 1: Numerical parameters (atomic units).

3 Results

3.1 Simulation 1: free Gaussian wavepacket

Setup. The initial state is a Gaussian with mean momentum k_0,

$$\psi(x, 0) = \frac{1}{(2\pi\sigma_0^2)^{1/4}} \exp\left[-\frac{(x - x_0)^2}{4\sigma_0^2}\right] e^{ik_0 x}, \tag{10}$$

with $x_0 = -8$, $\sigma_0 = 1$, $k_0 = 5$ and $V = 0$.

Analytic prediction. The width of a free Gaussian grows as

$$\sigma(t) = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2}\right)^2}, \tag{11}$$

while its centre moves at the group velocity $\langle x \rangle = x_0 + \hbar k_0 t/m$.

Physical interpretation. A Gaussian in x is a Gaussian in k, with width $1/(2\sigma_0)$. Each momentum component travels at its own velocity $\hbar k/m$, so faster components pull ahead and the packet spreads. This happens because the dispersion relation $\omega = \hbar k^2/2m$ is quadratic, not linear. At $t = 0$ the packet is a minimum-uncertainty state, $\Delta x\, \Delta p = \hbar/2$. Afterwards Δx grows while Δp stays fixed, because momentum is conserved for a free particle.

Results. Table 2 and Figure 1 compare the numerical and analytic widths. In the original box ($L = 20$) the agreement is exact to round-off up to $t = 3$. At $t = 4$ the error jumps to 1.6%. The packet's centre is then at $x = 12$ and its width is 2.24, so its leading tail crosses the periodic boundary at $x = 20$ and reappears at $x = -20$. The variance computed over $[-L, L)$ then counts that tail on the wrong side. This is an artefact of how the width is measured, not of the propagation: doubling the box to $L = 40$ removes it, and the error falls to 3×10^{-15}. The norm is conserved to better than 3×10^{-14} in both runs (Figure 1, right).

t	σ_{num} $(L = 20)$	σ_{an}	rel. error $(L = 20)$	rel. error $(L = 40)$
0	1.000000	1.000000	7×10^{-15}	7×10^{-15}
1	1.118034	1.118034	8×10^{-16}	1×10^{-15}
2	1.414214	1.414214	3×10^{-16}	3×10^{-15}
3	1.802776	1.802776	3×10^{-11}	2×10^{-15}
4	2.272507	2.236068	1.6×10^{-2}	3×10^{-15}

Table 2: Simulation 1: numerical versus analytic width. The $L = 20$ error at $t = 4$ comes from the packet tail wrapping around the periodic boundary; it disappears when the box is doubled.

3.2 Simulation 2: rectangular-barrier tunnelling

A rectangular barrier

$$V(x) = \begin{cases} V_0, & |x| \le a, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

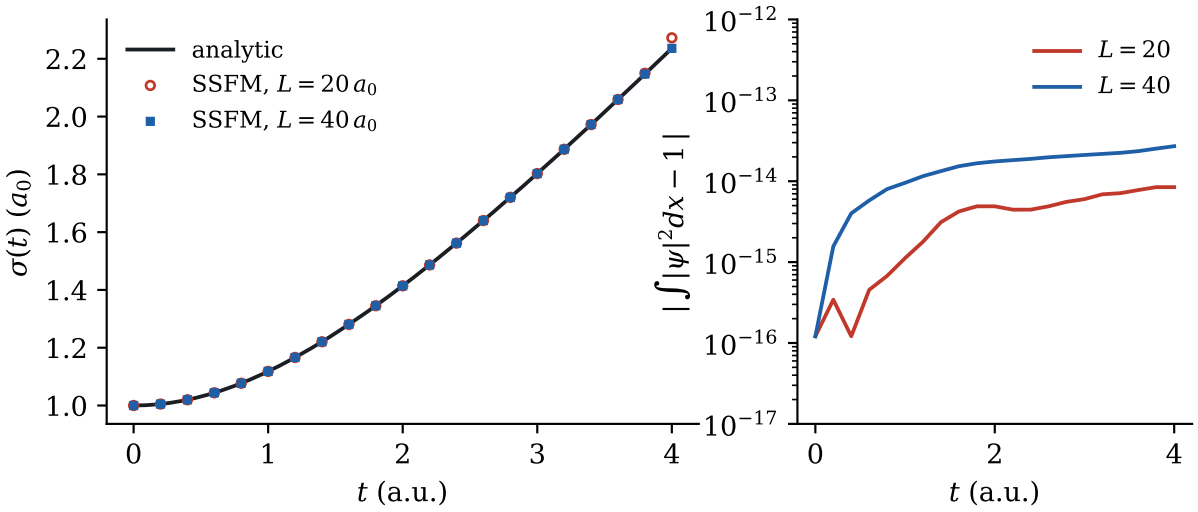


Figure 1: Left: width $\sigma(t)$ from equation (11) (line) and from the SSFM in the original box (circles) and a doubled box (squares). Right: deviation of the norm from 1, which stays at the level of round-off.

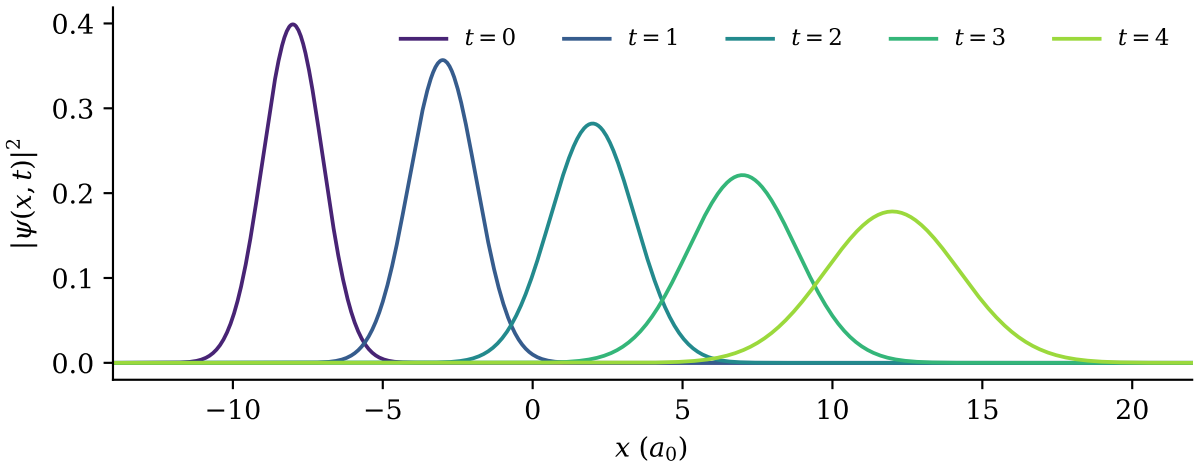


Figure 2: Probability density $|\psi(x,t)|^2$ at $t = 0, 1, 2, 3, 4$ ($L = 40$ run). The packet moves at $v_g = \hbar k_0/m = 5$ and broadens according to equation (11); its peak height falls as $1/\sigma(t)$.

is classically impenetrable for $E < V_0$, but quantum mechanics predicts a finite transmission. For a plane wave of energy E the exact result is [1]

$$T(E) = \left[1 + \frac{V_0^2 \sinh^2(2\kappa a)}{4E(V_0 - E)}\right]^{-1}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}, \tag{13}$$

with $\sinh \to \sin$ and $\kappa \to q = \sqrt{2m(E - V_0)}/\hbar$ for $E > V_0$.

A wavepacket is not a plane wave: it contains a spread of momenta, each transmitted with its own $T(k)$. The correct comparison is therefore the packet average

$$\bar{T} = \int |\tilde{\psi}(k)|^2 \, T\big(\hbar^2 k^2/2m\big)\, dk, \tag{14}$$

where $|\tilde{\psi}(k)|^2$ is a Gaussian of width $1/(2\sigma_0)$ about k_0. Numerically, T is the probability found at $x > a$ once the packet has fully scattered.

We used $V_0 = 1\,E_h$ and $2a = 1\,a_0$, with a wide packet ($\sigma_0 = 10$, so the momentum spread is only 0.05) and 15 values of k_0 between 0.6 and 2.0. Energies therefore run from $0.18\,E_h$, deep in

the tunnelling regime, to $2\,E_h$, well above the barrier. One discretisation detail matters here. A barrier drawn on a grid has an effective width equal to the number of cells inside it times Δx. On a first, careless grid this came out 1.1% too narrow, which raised T by up to 3% at low energy. Choosing $\Delta x = 1/32$ makes the barrier exactly 32 cells wide.

Figure 3 and Table 3 show the result. Over all 15 energies the SSFM transmission agrees with equation (14) to within 4.6×10^{-4} (at most 0.13% relative). It also reproduces the smooth crossover from tunnelling to over-barrier transmission at $E = V_0$.

E (E_h)	T_{SSFM}	$\bar{T}$, eq. (14)	T plane wave
0.180	0.1770	0.1771	0.1764
0.500	0.4199	0.4199	0.4200
0.980	0.6582	0.6586	0.6592
1.280	0.7550	0.7553	0.7559
2.000	0.8905	0.8908	0.8913

Table 3: Simulation 2: transmission at selected energies ($V_0 = 1$, $2a = 1$).

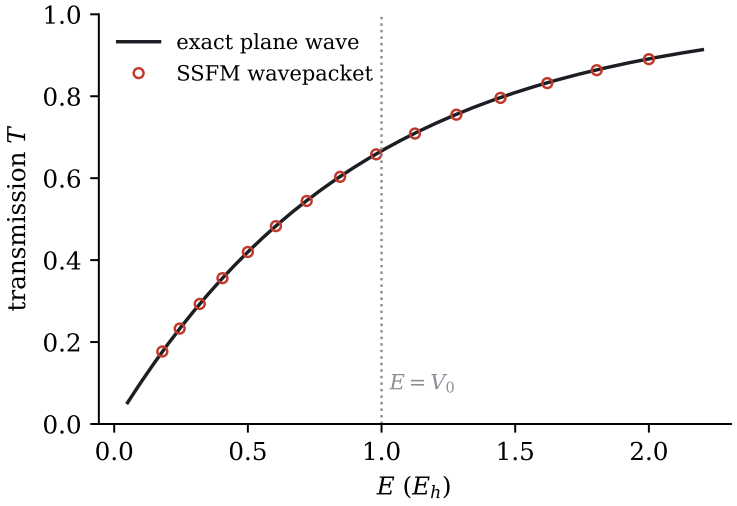


Figure 3: Transmission through a rectangular barrier ($V_0 = 1\,E_h$, width $1\,a_0$). Line: exact plane-wave result (13). Circles: SSFM wavepackets. Below $E = V_0$ all transmission is tunnelling.

3.3 Simulation 3: harmonic-oscillator coherent state

For $V(x) = \frac{1}{2}m\omega^2 x^2$, a Gaussian of width $\sigma = \sqrt{\hbar/2m\omega}$ displaced to $x = A$ is a coherent state. It keeps its shape exactly, and its centre follows the classical orbit

$$\langle x \rangle(t) = A \cos \omega t. \tag{15}$$

We used $\omega = 1$, $A = 3$ and ran for two full periods ($t = 4\pi$). The centre stays on the classical orbit to within $1.4 \times 10^{-4}\,a_0$. The width stays at $1/\sqrt{2}$ to within 9×10^{-6}, and the norm is conserved to 2×10^{-13} (Figure 4, left).

Because $V \neq 0$, this run also measures the order of the method. We propagated to $t = \pi$ with $\Delta t = \pi/20,\ \pi/40, \ldots, \pi/320$ and computed the L^2 distance to a reference run with $\Delta t = 10^{-4}$. Halving Δt cuts the error by a factor of 4.000. The fitted slopes are 2.000, 2.000, 2.000 and 2.000, confirming second-order accuracy exactly as equation (7) predicts (Figure 4, right).

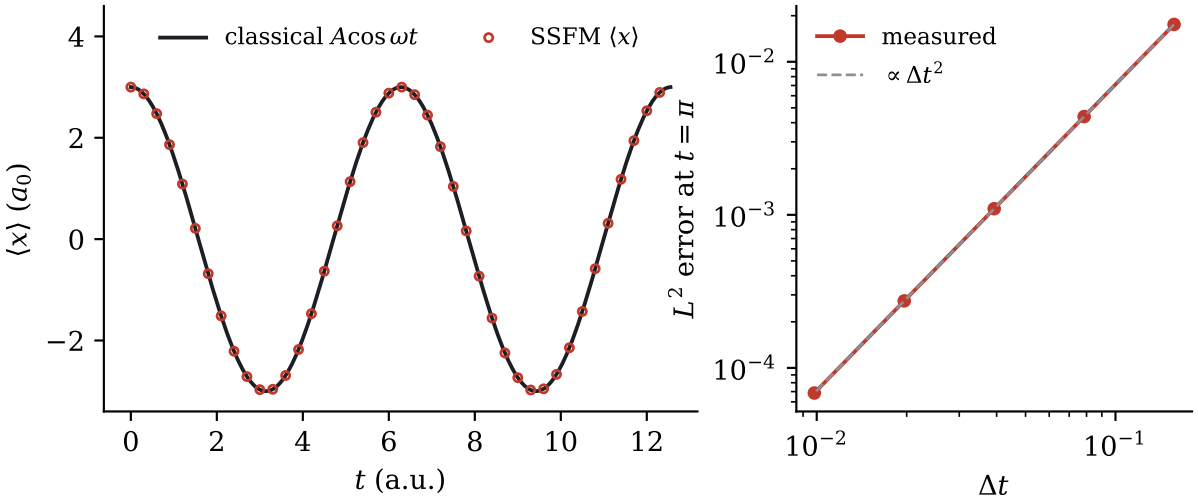


Figure 4: Left: centre of a coherent state (circles) against the classical orbit (15) (line) over two periods. Right: L^2 error at $t = \pi$ against time step; the dashed line has slope 2.

3.4 Simulation 4: double-slit interference

Interference needs two dimensions, so here the solver runs on a 1024×512 grid with a 2D FFT. A wall of height $V = 150\,E_h$ and thickness 0.4 stands at $x = 0$, pierced by two slits of width $w = 0.6$ centred at $y = \pm d/2$ with $d = 6$. A Gaussian packet ($\sigma_x = 2$, $\sigma_y = 6$, $k_0 = 6$, so $\lambda = 2\pi/k_0 = 1.047$) starts at $x = -10$ and moves towards the wall. Absorbing layers 4 units wide on all four edges remove the reflected and diffracted waves before they can wrap around the box.

Prediction. The two slits act as coherent sources with equal phase. In the far field, the transmitted amplitude is then proportional to $\cos(k_y d/2)$ times a single-slit envelope, so the bright fringes sit at

$$k_y = \frac{2\pi m}{d}, \qquad \text{equivalently} \qquad \sin\theta_m = \frac{m\lambda}{d}. \tag{16}$$

The form in k_y is the cleaner test: it does not depend on the particle's energy, so the packet's momentum spread cannot blur it. We therefore Fourier transform the transmitted part of the wavefunction ($x > 1$) at $t = 8$ and sum over k_x.

Results. Figure 5 shows the density and the transverse-momentum distribution. On the grid, the slit centres resolve to a separation of $d = 6.094$, which predicts $k_y = 1.031\,m$. The measured peaks for $m = 1$ to 5 fall at 1.021, 2.042, 3.089, 4.084 and 5.131. Every peak is within 1% of the prediction, and every one is within a single bin of the momentum grid ($\Delta k_y = 0.026$). About 15% of the probability passes through the slits.

3.5 Simulation 5: step potential

For a step $V(x) = V_0\,\theta(x)$ with $E > V_0$, a classical particle always crosses. Quantum mechanics predicts partial reflection:

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2, \qquad T = \frac{4k_1 k_2}{(k_1 + k_2)^2}, \tag{17}$$

with $k_1 = \sqrt{2mE}/\hbar$ and $k_2 = \sqrt{2m(E - V_0)}/\hbar$. As before, we compare against the packet average of $R(k)$. Components with $E < V_0$, present only in the tail of the distribution near threshold, are totally reflected.

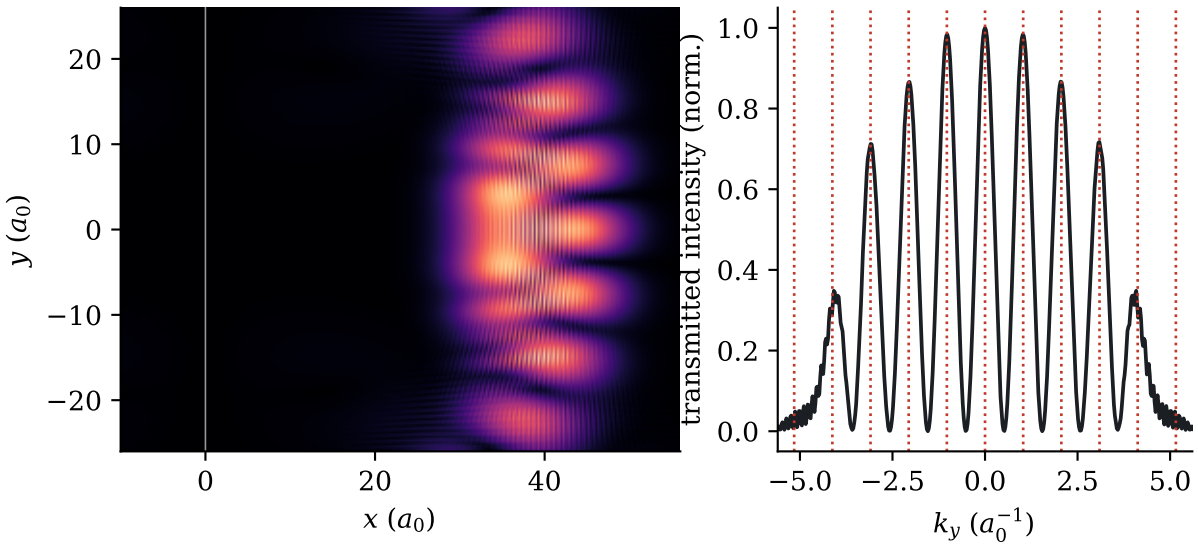


Figure 5: Left: $|\psi|$ at $t = 9$, after the packet has passed the slits at $x = 0$ (white line); interference fringes are visible downstream. Right: transverse-momentum distribution of the transmitted wave. Dotted lines mark the predicted bright fringes $k_y = 2\pi m/d$.

With $V_0 = 1\,E_h$ and five energies from 1.28 to $4.5\,E_h$, the measured reflection agrees with the packet average to within 8×10^{-5} (Table 4, Figure 6). The comparison is sensitive near threshold. At $E = 1.28$ the plane-wave value is 0.1315, while both the simulation and the packet average give 0.1400. The difference comes entirely from the slow momentum components that cannot climb the step, and the simulation captures it.

E (E_h)	R_{SSFM}	$\bar{R}$ (packet)	R plane wave
1.280	0.14001	0.13994	0.13153
1.620	0.05689	0.05682	0.05551
2.000	0.02990	0.02984	0.02944
3.125	0.00934	0.00930	0.00924
4.500	0.00398	0.00395	0.00394

Table 4: Simulation 5: reflection probability above a step of height $V_0 = 1\,E_h$.

4 Discussion

4.1 The Ehrenfest theorem

Ehrenfest's theorem states that expectation values obey classical equations of motion:

$$\frac{\mathrm{d}\langle x\rangle}{\mathrm{d}t} = \frac{\langle p\rangle}{m}, \qquad \frac{\mathrm{d}\langle p\rangle}{\mathrm{d}t} = -\left\langle \frac{\mathrm{d}V}{\mathrm{d}x} \right\rangle . \tag{18}$$

This does not mean the particle follows a trajectory. It means the centre of the packet does, exactly so when dV/dx is linear in x, as for the harmonic oscillator. Simulation 3 shows this directly. The barrier and step show the opposite case: there the potential changes over a distance much shorter than the packet, and the packet splits into transmitted and reflected parts that no single classical trajectory describes.

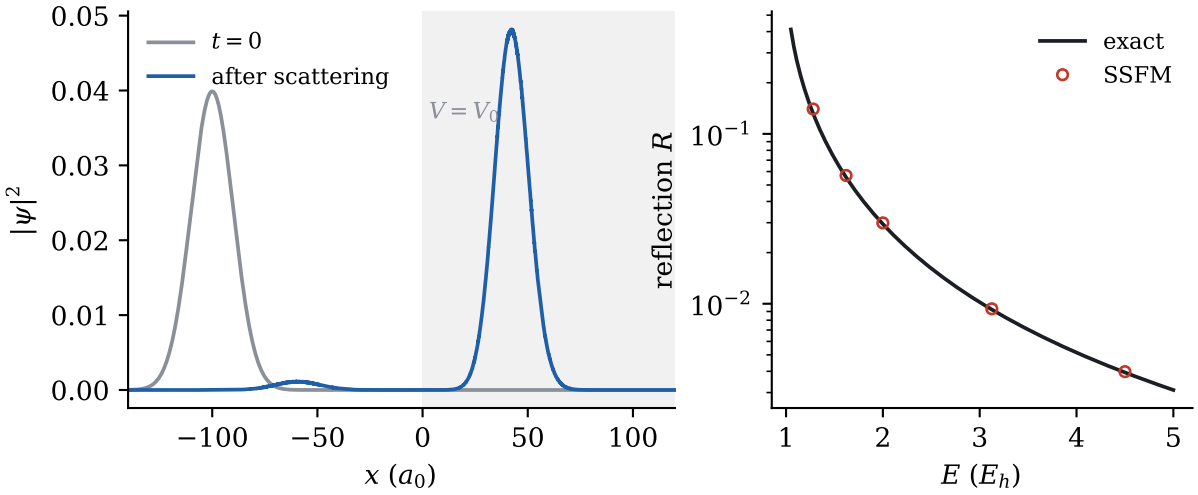


Figure 6: Left: a packet with $E = 2E_h$ before and after meeting the step (shaded); a small reflected packet returns to the left. Right: reflection probability against energy, exact (17) (line) and SSFM (circles).

4.2 When does the SSFM fail?

1. **Aliasing.** Wavenumbers above $k_{\max} = \pi/\Delta x$ wrap around in the FFT. This happens when a packet meets a very steep potential or the grid is too coarse. The remedy is a larger N.

2. **Periodic boundaries.** The FFT treats the domain as a ring. Simulation 1 shows the consequence: a packet tail reaching the edge corrupts the measured width by 1.6%. The remedies are a larger box or an absorbing layer, as in Simulation 4.

3. **Grid representation of sharp potentials.** A step or barrier can only begin and end on grid points. Simulation 2 shows that a 1% error in the effective barrier width becomes a 3% error in the tunnelling probability. Sharp features should be aligned with the grid.

4. **Time step.** The splitting error scales as Δt^2 times the size of the commutator $[\hat{T}, \hat{V}]$. It is therefore largest where the potential varies rapidly, and absent when $V = 0$.

4.3 Connection to broader physics

The SSFM is used well beyond textbook problems:

- **Quantum and nonlinear optics:** pulse propagation in dispersive fibres, governed by the nonlinear Schrödinger equation;
- **Strong-field physics:** ionisation of atoms by intense laser pulses;
- **Bose–Einstein condensates:** the Gross–Pitaevskii equation, which has the same structure as the TDSE plus a nonlinear term.

The same operator-splitting idea gives the symplectic Verlet integrator of classical molecular dynamics and appears in lattice field theory, and the BCH formula behind it is central to the theory of Lie groups.

5 Conclusion

We implemented and validated a split-step Fourier solver for the time-dependent Schrödinger equation in one and two dimensions. Tested against five problems with known answers, it reproduces:

1. free-particle dispersion to 10^{-14}, once the box is large enough;
2. tunnelling and over-barrier transmission to 5×10^{-4} across a factor of 11 in energy;
3. coherent-state motion to $1.4 \times 10^{-4}\, a_0$, with exactly second-order convergence in Δt;
4. double-slit fringe positions to within one momentum-grid bin;
5. above-step reflection to 8×10^{-5}, including the near-threshold correction from the packet's momentum spread.

The norm is conserved to round-off in every closed-box run. The errors that did appear (boundary wrap-around and the grid width of a sharp barrier) were traced to their causes and removed, and the discussion records them as practical lessons. Natural next steps are time-dependent potentials, such as an atom in a laser field, and the nonlinear Gross–Pitaevskii equation.

A Python implementation

The core solver, shared by all five simulations:

Listing 1: The SSFM propagator, diagnostics and analytic results (`ssfm.py`).

```python
"""Split-step Fourier solver for the time-dependent Schrodinger equation.

Atomic units throughout (hbar = m = 1). Shared by all five simulations.
"""
import numpy as np


def grid_1d(N, L):
    """Periodic grid on [-L, L) and its angular wavenumbers."""
    x = np.linspace(-L, L, N, endpoint=False)
    dx = x[1] - x[0]
    k = 2 * np.pi * np.fft.fftfreq(N, d=dx)
    return x, dx, k


def gaussian(x, x0, sigma0, k0):
    """Normalised Gaussian wavepacket with mean momentum k0."""
    psi = np.exp(-(x - x0) ** 2 / (4 * sigma0 ** 2) + 1j * k0 * x)
    dx = x[1] - x[0]
    return psi / np.sqrt(np.sum(np.abs(psi) ** 2) * dx)


class SSFM:
    """Strang-split propagator: half potential kick, full kinetic kick, half potential kick.

    V may be complex: a negative imaginary part acts as an absorbing layer.
    Works for 1D or 2D arrays (k2 must be |k|^2 on the same grid).
    """

    def __init__(self, V, k2, dt, hbar=1.0, m=1.0):
        self.half_V = np.exp(-1j * V * dt / (2 * hbar))
        self.full_T = np.exp(-1j * hbar * k2 * dt / (2 * m))
        self.dt = dt

    def step(self, psi, n=1):
        fft, ifft = (np.fft.fft, np.fft.ifft) if psi.ndim == 1 else (np.fft.fft2, np.fft.ifft2)
        for _ in range(n):
            psi = self.half_V * psi
            psi = ifft(self.full_T * fft(psi))
            psi = self.half_V * psi
```

```
        return psi


def moments(psi, x):
    """Norm, <x> and width sigma from |psi|^2 (rectangle rule, exact on a periodic grid)."""
    dx = x[1] - x[0]
    p = np.abs(psi) ** 2
    norm = np.sum(p) * dx
```

References

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